ANALYTICAL CHEMISTRY CUMULATIVE EXAM

March 2, 1991

In 1985 two Danish scientists, published a paper entitled: "Piezoelectric Crystal Detector for the Monitoring of Ozone in Working Environments".[1] Ozone is a common hazard in industrial environments were electric arcs from motors or welding equipment produces O_3 from O_2.

The principle of the piezoelectric detector is that the frequency of a coated crystal is decreased by adsorption of gaseous ozone on the crystal surface. To make the crystal surface selective for ozone, they coated the crystal with a thin layer of 1,4-polybutadiene, which grabs O_3 out of the air. The crystal only responds to the mass of ozone on the surface at any one time according to the equation:

$$\Delta F = -K(\Delta M)$$

When no ozone is present ΔM is zero and ΔF is zero.

There are basically two types of detector surfaces, those in which the analyte is in a reversible equilibrium between the air and the polymer coating, and those polymers which adsorb the analyte irreversibly, never to surrender it.

1. Give an argument which justifies that with reversible reactions the frequency ΔF of the detector depends upon the chemical equilibrium between the concentration of gas in the ambient air C, and gas bound to the polymer, according to:

$$\Delta F = -K_1 C$$

2. Give an argument which justifies that with irreversible reactions the frequency ΔF of the detector depends upon the concentration in the air C, and the time of exposure Δt, to the polymer, according to:

$$\Delta F = -K_2 C \Delta t$$

3. What determines the sensitivity of the technique in the reversible case? The irreversible case?

4. It turns out that for the ozone/butadiene system the detection mode is irreversible. How would you determine what period of time you would expose the sensor to the environment?

5. What are the advantages or disadvantages between the reversible and irreversible sensors?

1. Fog, H.M.; Rietz, B., Anal. Chem., 57, 2634-2638, 1985.